

# **Mobile and Network Forensics**

## **Activity- 3**

### **Call Detail Records Analysis**

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**Introduction:**

In digital forensics, call detail records (CDRs) are crucial data sources since they record call and message details like time, duration, and involved contacts. Understanding communication patterns and spotting strange conduct can be made easier by analyzing this data.

Using information gathered from the Airtel e-prebill service by submitting a request to 121, a CDR analysis system is created in this project. The technology creates insights like commonly contacted numbers, active hours, and daily communication trends by processing call and SMS information.

Additionally, a simple system for detecting suspicious activity is put in place based on parameters like call frequency, duration, and late-night activity. Graphs and an interactive online dashboard made with Flask are used to display the results.

### **Problem Statement:**

Large volumes of communication data are present in Call Detail Records (CDRs), making it challenging to manually examine and extract pertinent information. When working with raw data, it becomes difficult and time-consuming to find patterns like frequent interactions, peak activity hours, or suspicious conduct.

A system that can automatically analyze CDR data and provide insightful information in an intelligible manner is required. Furthermore, it is crucial for forensic analysis to identify unusual communication patterns based on variables like call frequency, duration, and time.

By creating a system that evaluates CDR data, displays communication trends, and uses an interactive dashboard to identify possibly suspicious activities, this project tackles these issues.

### **Objectives**

- To collect and analyze CDR data obtained from the Airtel e-prebill service
- To process call and SMS records using Python
- To identify frequently contacted numbers
- To analyze communication patterns based on time and date

- To visualize data using graphs such as active hours and daily trends
- To detect potentially suspicious activity based on predefined factors
- To develop a web-based dashboard using Flask for easy analysis and visualization

## **Tools Used and System Requirements**

This project requires a basic knowledge of data analysis and programming. The technology and tools listed below were employed:

### **Programming Language:**

#### **Python**

→ utilized in backend logic, data processing, and analysis.

### **Libraries Used:**

#### **Pandas**

→ utilized for data cleansing, analysis, and CSV file reading.

**Matplotlib** used to create graphs like:

- Leading contacts
- Hours of operation
- Everyday patterns
- SMS activity

#### **Flasks**

→ used to build a straightforward web-based dashboard for outcomes. **Folium**

→ used to create an interactive map for location visualization.

### **Frontend:**

#### **HTML**

→ Used to structure and display the dashboard interface.

## **Methodology**

The project follows a practical approach starting from real data collection to analysis and visualization of Call Detail Records (CDR).

### **1. Data Collection**

The CDR data was collected using the Airtel e-prebill service. A request message was sent to 121 in the format:

EPREBILL <Month> <Email ID>

After sending the request, an email was received containing the call and SMS records in a password-protected PDF file.

### **2. Data Extraction**

The received PDF file was unlocked using the provided password. Relevant data such as:

- Voice call details
- SMS records were manually extracted and converted into structured CSV files:
- cdr\_voice.csv
- cdr\_sms.csv

### **3. Data Preparation**

The extracted data was organized into CSV format to make it suitable for processing. This included:

- Arranging columns like Number, Date, Time, and Duration
- Ensuring consistent formatting for analysis

### **4. Data Analysis Using Python**

A Python script was developed to process and analyze the CSV files.

File name :

- cdr\_script.py

The script performs:

- Reading datasets using Pandas
- Extracting time (hour) and date information
- Calculating total calls, SMS, and duration

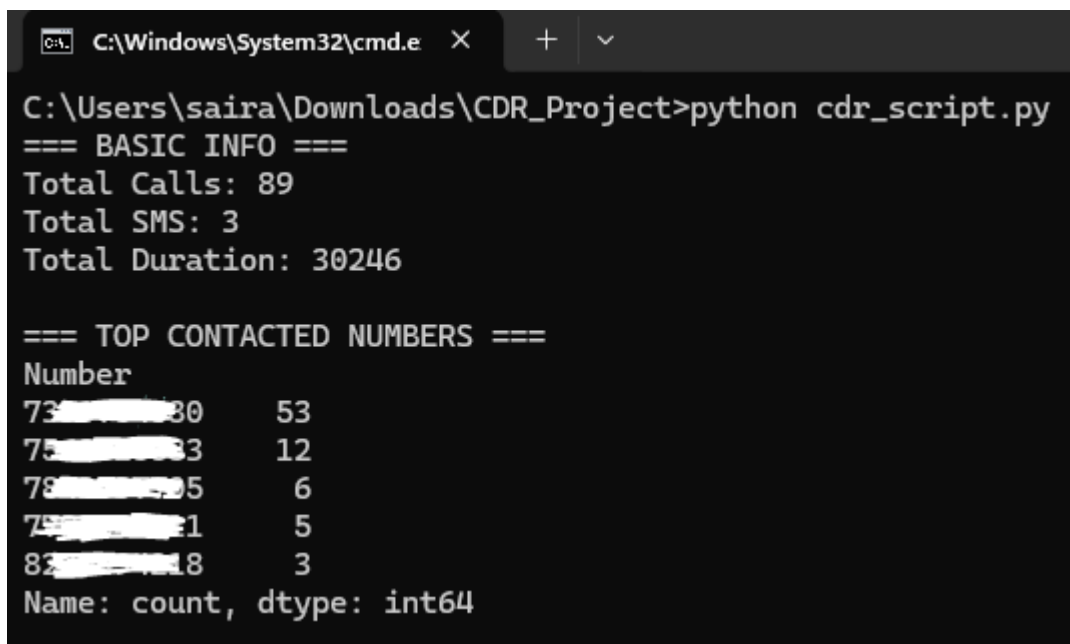
## 5. Execution and Output Generation

The Python script (cdr\_script.py) is executed using Command Prompt:

“python cdr\_script.py”

On execution, the script processes the CDR data and generates all output graphs automatically. These graphs are saved in a separate folder named graphs.

The interactive map (map.html) is also generated and stored in the same graphs folder.



```

C:\Windows\System32\cmd.e  X  +  v

C:\Users\saira\Downloads\CDR_Project>python cdr_script.py
=== BASIC INFO ===
Total Calls: 89
Total SMS: 3
Total Duration: 30246

=== TOP CONTACTED NUMBERS ===
Number
73[REDACTED]30      53
75[REDACTED]33      12
78[REDACTED]05       6
79[REDACTED]1        5
82[REDACTED]18       3
Name: count, dtype: int64

```

## 5. Visualization

Graphs were generated using Matplotlib to represent:

- Frequently contacted numbers
- Active call hours
- Daily call trends
- SMS activity patterns

Graphs were saved in folder name:

- Graphs

These visualizations help in understanding communication behavior clearly.

## 6. Telecom Data Simulation

Basic telecom-related details such as operator and cell information were generated based on number prefixes to simulate real-world telecom analysis.

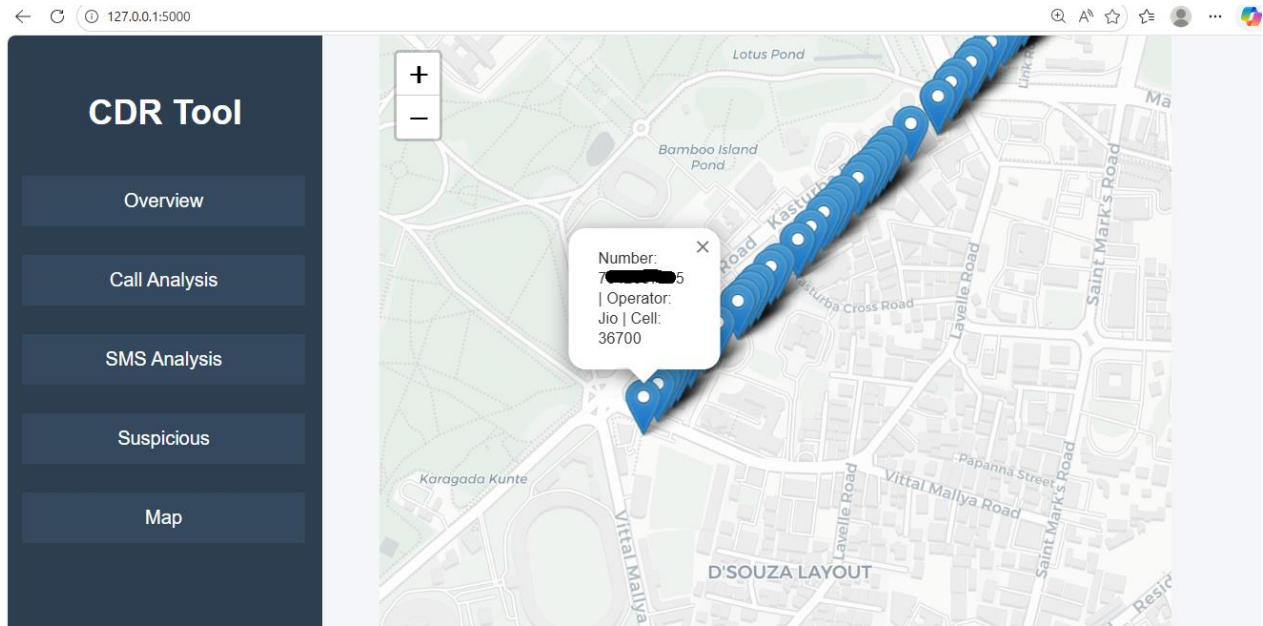
```
=== TELECOM DETAILS ===
```

|   | Number       | MCC | MNC | Operator | Cell_ID | LAC  | TAC   |
|---|--------------|-----|-----|----------|---------|------|-------|
| 0 | 73-██████-30 | 404 | 45  | Airtel   | 98270   | 1819 | 80612 |
| 1 | 72-██████-30 | 404 | 45  | Airtel   | 86804   | 1922 | 84712 |
| 2 | 73-██████-30 | 404 | 45  | Airtel   | 87919   | 4136 | 77847 |
| 3 | 72-██████-10 | 404 | 00  | Unknown  | 40081   | 5661 | 88189 |
| 4 | 72-██████-05 | 405 | 51  | Jio      | 29693   | 2528 | 31530 |
| 5 | 72-██████-05 | 405 | 51  | Jio      | 36700   | 6542 | 17329 |
| 6 | 72-██████-48 | 404 | 00  | Unknown  | 70605   | 2668 | 72761 |
| 7 | 72-██████-05 | 405 | 51  | Jio      | 10074   | 4240 | 83329 |
| 8 | 72-██████-33 | 404 | 10  | Airtel   | 97471   | 1881 | 28924 |
| 9 | 72-██████-05 | 405 | 51  | Jio      | 45575   | 8768 | 30299 |

Map saved → graphs/map.html

## 7. Location Mapping

Using Folium, an interactive map was created to display approximate locations based on generated cell IDs. This helps visualize communication activity geographically.



## 8. Suspicious Activity Detection

A simple scoring method was applied to identify unusual communication patterns based on:

- High call frequency
- Long duration calls
- Late-night activity
- Repeated short calls

```

=== SUSPICIOUS ANALYSIS ===

Top Suspicious Numbers:

```

| Number         | Frequency | LongCalls | LateNight | ShortCalls | Score |
|----------------|-----------|-----------|-----------|------------|-------|
| 73[Redacted]30 | 53        | 10.0      | 13.0      | 6.0        | 168.0 |
| 75[Redacted]3  | 12        | 0.0       | 0.0       | 0.0        | 24.0  |
| 78[Redacted]5  | 6         | 0.0       | 0.0       | 3.0        | 15.0  |
| 77[Redacted]11 | 5         | 0.0       | 0.0       | 2.0        | 12.0  |
| 82[Redacted]18 | 3         | 0.0       | 0.0       | 1.0        | 7.0   |

Contacts with higher scores are considered potentially suspicious.

## 9. Result Presentation

The final output is displayed through:

- Graphs generated in the “graphs” folder

| Name             | Date modified    | Type                | Size  |
|------------------|------------------|---------------------|-------|
| ▼ Yesterday      |                  |                     |       |
| map.html         | 30-04-2026 22:26 | Microsoft Edge H... | 72 KB |
| suspicious.png   | 30-04-2026 22:26 | PNG File            | 21 KB |
| sms_hours.png    | 30-04-2026 22:26 | PNG File            | 11 KB |
| daily_trend.png  | 30-04-2026 22:26 | PNG File            | 26 KB |
| top_sms.png      | 30-04-2026 22:26 | PNG File            | 15 KB |
| active_hours.png | 30-04-2026 22:26 | PNG File            | 13 KB |
| top_contacts.png | 30-04-2026 22:26 | PNG File            | 19 KB |

## 10. Flask Application (UI\_script.py)

The web interface of the system is developed using Flask. The script connects the analyzed data with the frontend and displays summary information such as total calls, SMS, and duration.

### Key functions:

- Loads CDR datasets
- Calculates basic statistics

Script used in cmd : “python UI\_script.py”

```
C:\Users\saira\Downloads\CDR_Project>python UI_script.py
* Serving Flask app 'UI_script'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 355-417-117
127.0.0.1 - - [30/Apr/2026 23:05:32] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [30/Apr/2026 23:05:40] "GET /graphs/sms_hours.png HTTP/1.1" 200 -
127.0.0.1 - - [30/Apr/2026 23:05:41] "GET /graphs/top_contacts.png HTTP/1.1" 200 -
127.0.0.1 - - [30/Apr/2026 23:05:41] "GET /graphs/top_sms.png HTTP/1.1" 200 -
127.0.0.1 - - [30/Apr/2026 23:05:41] "GET /graphs/daily_trend.png HTTP/1.1" 200 -
127.0.0.1 - - [30/Apr/2026 23:05:41] "GET /graphs/active_hours.png HTTP/1.1" 200 -
127.0.0.1 - - [30/Apr/2026 23:05:41] "GET /graphs/suspicious.png HTTP/1.1" 200 -
127.0.0.1 - - [30/Apr/2026 23:05:52] "GET /graphs/map.html HTTP/1.1" 200 -
```

## 11. User Interface (web\_page.html)

The frontend of the system is developed using HTML. It provides a simple dashboard layout with multiple sections such as overview, call analysis, SMS analysis, suspicious activity, and map visualization.

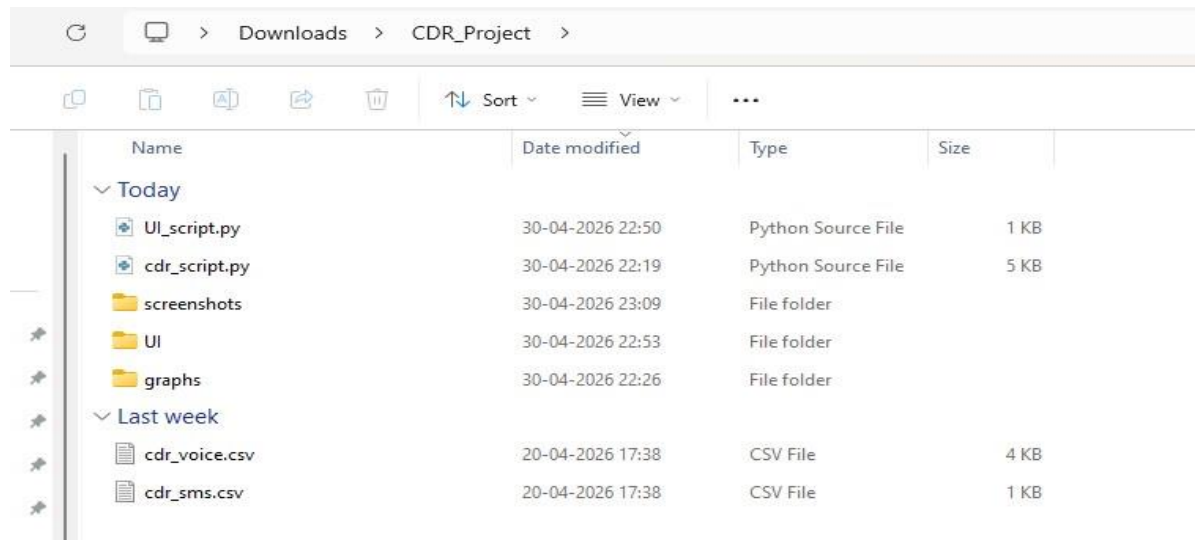
### Key features:

- Sidebar navigation
- Graph display using images
- Map integration using iframe



The interface is accessed through a web browser by running the Flask application and entering the local server address (<http://127.0.0.1:5000>), where the dashboard is displayed.

### The final folder:



## Challenges faced

- Integrating Flask backend with HTML frontend
- Displaying the map in the UI (path and iframe issues)
- Generating telecom information, as real telecom data is not available in e-prebill and had to be simulated based on number prefixes
- Designing logic for suspicious activity detection

## Results

Through the analysis of CDR data, the system produced graphs that provided valuable information. Communication trends like frequently contacted numbers and active hours were detected.

While daily statistics revealed changes over time, call and SMS data aided in understanding usage habits. Contacts exhibiting odd patterns based on frequency, duration, and time were identified by the suspicious activity detector.

The analysis was further contextualized by the telecom data simulation and map visualization.

All things considered, the technology successfully transforms unprocessed CDR data into clear and intelligible insights.

## Conclusion

The project effectively created a CDR analysis system that analyzes and displays call and SMS data to improve comprehension of communication trends. Large dataset analysis is made easier by the system's use of Python, data analysis modules, and an online interface.

Important insights like frequently contacted numbers, active communication hours, and suspicious activities can be identified by the system. Graphs and an interactive dashboard enhance usability and facilitate interpretation of the information.

The experiment shows how fundamental data analysis methods may be successfully used in digital forensics, even though actual telecom-level data was unavailable and had to be simulated.

In summary, this study emphasizes the value of automation and visualization in forensic investigations and offers a useful method for analyzing CDR data.

## Project Folder



CDR\_Project.zip